



THE UNIVERSITY OF
SYDNEY

Room Number _____
Seat Number _____
Student Number

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ANONYMOUSLY MARKED

(Please do not write your name on this exam paper)

SAMPLE EXAM PAPER

This paper is not to be removed from the exam venue

Electrical and Information Engineering

EXAMINATION

SAMPLE EXAM

ELEC3607/ELEC9607 Embedded Systems

For Examiner Use Only

EXAM WRITING TIME: 2 hours
READING TIME: 10 minutes

EXAM CONDITIONS:

This is a CLOSED book exam - no material permitted

This is a sample which has not been thoroughly checked, no solutions will be provided.

MATERIALS PERMITTED IN THE EXAM VENUE:

(No electronic aids are permitted e.g. laptops, phones)

Calculator - non-programmable

MATERIALS TO BE SUPPLIED TO STUDENTS:

1 x 12-page answer book

INSTRUCTIONS TO STUDENTS:

Please tick the box to confirm that your examination paper is complete. ☐

Q	Mark
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Total _____

Question 1 – Peripherals (25 marks)

Please refer to extracts from the data sheet for the Renesas 1339 I2C bus real-time clock at the end of this examination script.

- a. (10 marks) Explain the sequence of register reads and writes needed to set the time to 0123 in 24 hour mode on the 1/2/2023.

Your answer should try to minimise the number of register accesses to configure this.

- b. (10 marks) Using a figure, give an example of an I2C write transaction from the Master that is not acknowledged by the Slave.
- c. (5 marks) Draw a schematic diagram showing how you would interface a Renesas 1339-2 chip to a microcontroller via the I2C bus, with super capacitor trickle charging of 2mA. Show all the external components required for this interface and show your calculations.

Question 2 – Interfacing (25 marks)

- a. (5 marks) Explain how one could interface multiple Renesas 1339 devices to a single microcontroller.
- b. (5 marks) Explain why bypass capacitors are needed even for purely digital designs. What type of capacitor is suitable for high speed bypassing?
- c. (5 marks) Give a situation in which an active filter would be preferred over a passive filter?
- d. (5 marks) Optoisolators are often integrated into solid state relays. Explain their purpose using an example.
- e. (5 marks) USB uses a tree network topology and master/slave protocol. Explain the meaning of these terms and how they are implemented in practice.

Question 3 – High Speed Systems (25 marks)

- a. (5 marks) Given that characteristic impedance is calculated as

$$Z_o = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

where

R is the **resistance** per unit length, considering the two conductors to be **in series**,

L is the **inductance** per unit length,

G is the **conductance** of the dielectric per unit length,

C is the **capacitance** per unit length,

j is the **imaginary unit**, and

ω is the **angular frequency**.

in a microstrip transmission line structure, is the characteristic impedance increased or decreased as you increase the width of the signal trace? Explain your answer.

- b. (5 marks) Suppose I have a printed circuit board with solid ground plane. How should I connect the analog and digital grounds of a microcontroller which has both pins on the same package?
- c. (5 marks) Using a figure, explain the concept of ground bounce and how it can be minimised.
- d. (10 marks) The printed circuit board below has split analog and digital ground planes that are isolated from each other. A high speed signal must be sent between them using the horizontal trace shown in yellow. The board is 4 layers: (1) top signal layer, (2) middle split analog/digital ground, (3) 3.3V and (4) bottom signal layer. How would you recommend that the digital and analog ground planes be connected together using a signal layer?

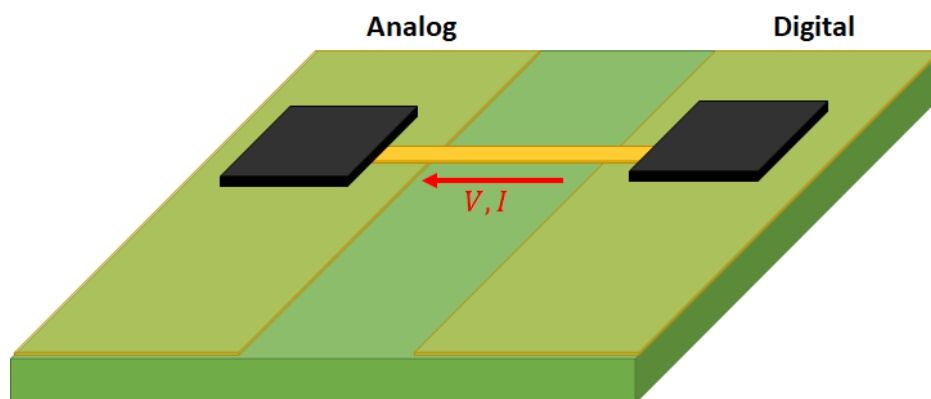


Figure: <https://www.nwengineeringllc.com/article/should-you-split-ground-planes-in-mixed-signal-pcb-design.php>

Question 4 – Operating Systems and Programming

- a. (5 marks) Explain the role of an operating system kernel.
- b. (5 marks) In a typical Linux boot sequence, what is the role of the Secondary Boot Loader?
- c. (5 marks) In ARM assembly language, write a loop that places the sum of the first 100 numbers in register w1. Use register w2 as the loop variable.
- d. (10 marks) Annotate each line of the following ARM assembly language program and explain what the function implemented computes.

```
mystery:
    sub    sp, sp, #16
    str    w0, [sp, 12]
    str    w1, [sp, 8]
    ldr    w0, [sp, 12]
    mul    w1, w0, w0
    ldr    w0, [sp, 8]
    mul    w0, w0, w0
    add    w0, w1, w0
    add    sp, sp, 16
    ret
```

- In the examination I would have attached an excerpt of this datasheet:

<https://www.renesas.com/us/en/document/dst/1339-datasheet?r=69011>

END OF EXAMINATION